



Influence of optical aperture sizes on aero-optical effects induced by supersonic turbulent boundary layers

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Abstract: With the increase of flight speed, aero-optical effects induced by the turbulent boundary layer near the optical window become increasingly significant. The density field of the supersonic (Mach 3.0) turbulent boundary layer (SPTBL) was measured by nano-tracer-based planar laser scattering technique, and the optical path difference (*OPD*) was obtained through ray-tracing method. The influence of the optical aperture size on the aero-optical effects of SPTBL was studied in detail, and the underlying mechanisms were analyzed from the perspective of the turbulent structure scales. The influence of the optical aperture on the aero-optical effects is mainly due to turbulent structures with different scales. The beam center jitter ($s_{\bar{x}}$) and offset ($\bar{x}$) are mainly caused by turbulent structures larger than the optical aperture size, while the beam spread about the center ($\overline{x'^2}$) is mainly caused by turbulent structures smaller than the optical aperture size. With the increase of optical aperture size, the proportion of turbulent structures larger than the optical aperture size decreases, which can suppress the beam jitter and the beam offset. Meanwhile, since the beam spread is primarily induced by small-scale turbulent structures with relatively strong density fluctuation intensity, the spread increases rapidly to its peak and then gradually stabilizes as the optical aperture size grows.

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1. Introduction

The aero-optical effect is an important factor affecting the performance of airborne optical systems. With the development of high-speed aircraft, the aero-optical effects caused by compressible turbulent flow in high-speed flow field has attracted more attention [1–5]. The aero-optical effects are caused by the density fluctuation of the flow field at the source. The flow field can be divided into atmospheric region and turbulent region. The density fluctuation in the atmospheric region is mainly dominated by the low-frequency fluctuation of temperature, and its mechanism and correction have been adequately studied [6,2]. And the aero-optical effects caused by the turbulent boundary layer (TBL) near the optical window became a hot research topic in the past decade, especially under the conditions of high-speed (from transonic to hypersonic) flow fields [4,7–9]. The aberration is usually characterized by optical path difference (*OPD*), and the intensity of aero-optical effect is characterized by OPD_{rms} , which is the root mean square (RMS) of *OPD* within the optical aperture.

The research on the aero-optics of TBL is mainly carried out by numerical simulation and experimental research. In terms of numerical simulations, due to the influence of high Reynolds number and compressibility under supersonic conditions, higher computational power is required compared to subsonic conditions. Early numerical studies were mostly unable to accurately

reflect the aero-optical effects of the flows due to these challenges [10–12], and therefore, early research primarily relied on experiments. In terms of experimental research, Wyckham and Smits used a Shack-Hartmann wavefront sensor to study the aero-optics of transonic and hypersonic boundary layers, and they proposed a scaling law [13]. Gordeyev *et al.* conducted a series of studies of subsonic and SPTBL using Malley probes and Shack-Hartmann wavefront sensors [8,14,15]. They verified and improved the scaling law proposed by Wyckham. The scaling law is used to validate the experimental data in this paper.

In existing studies on aero-optical effects of TBL, researchers have realized that the optical aperture has influence on the aero-optical law. In the case of subsonic flow over a flat-window turret, the experimental results of Porter *et al.* showed that the large-aperture approximation could lead to an underestimation of the Strehl ratio of practical optical systems with finite apertures [16]. Gordeyev's experimental results of SPTBL showed that, due to the presence of large-scale structures, the optical aperture size should be larger than 10δ to contain complete aero-optical distortions [14]. In a theoretical manner, it enables us to comprehend the aero-optical distortions associated with varying optical aperture sizes. Via considering a model index-of-refraction spectrum, Mathews *et al.* studied the relationship between OPD_{rms} and turbulent scales with different optical aperture sizes. With the reduce of the optical aperture size, OPD_{rms} tend to be decrease [17].

The influence of optical aperture sizes on aero-optical effects induced by SPTBL is closely related to turbulent scales. Wyckham and Smits' research revealed that the large-scale structures with the scale as the TBL thickness (δ) play a dominant role in aero-optical distortion [13]. Gordyev and Jumper found that the dominant large-scale structures of aero-optical distortion were pseudo-periodic structures with typical size of 1.5δ [14]. Based on the refractive index field of SPTBL measured by nano-tracer-based planar laser scattering (NPLS) technique, Gao *et al.* used the pre-multiplied power spectrum method to study the turbulent structures with different scales [18]. Ding *et al.* spatially filtered the refractive index field of SPTBL to study the influence of spatial resolution on OPD_{rms} [19]. In further, based on curvelet transform method, Ding *et al.* extracted different flow scales to study the influence of different scale turbulent structures on beam deflection diffusion, and found that the small-scale turbulent structure had a more significant influence on beam spread [20,21].

In this paper, the influence of optical aperture sizes on aero-optical effects and its underlying mechanisms were investigated. The rest of the paper was arranged as follows. Section 2 introduced the experimental conditions, including the experimental equipment, TBL, density field measurement based on NPLS, ray-tracing method and curvelet transform decomposition method briefly. Section 3 analyzed the influence of optical aperture sizes on aero-optical distortion. Finally, the conclusions of this paper were summarized in Section 4.

2. Experiment and analysis methods

2.1. Wind tunnel

Figure 1 was the schematic diagram of supersonic wind tunnel used in this paper. The dimension of the test section was $250\text{mm} \times 100\text{mm} \times 120\text{mm}$. The optical side window was made of optical glass, which could meet the requirements of optical measurement. The optical glass at the bottom window could effectively reduce the impact of reflection on the measurement results of NPLS technique. More detailed parameters of the wind tunnel were shown in Table 1.

Table 1. Operating parameters of the supersonic wind tunnel

Ma	p_0	T_0	ρ_∞	U_∞	Re	Turbulence Intensity	Running Time
3.0	1atm	300K	0.089kg/m^3	622.5m/s	$7.46 \times 10^6\text{m}^{-1}$	<0.5%	>10s

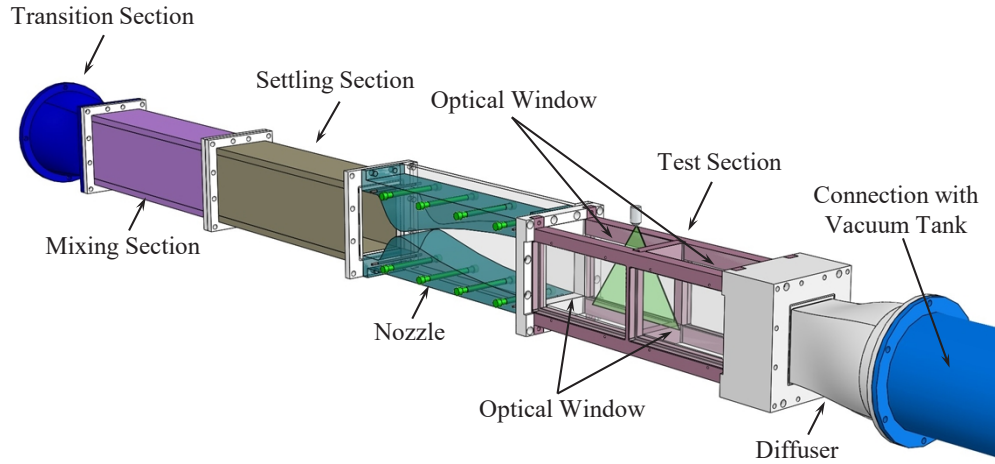


Fig. 1. Schematic diagram of the supersonic wind tunnel.

2.2. Experimental SPTBL

In this paper, the lower wall boundary layer at 180mm downstream of the wind tunnel was selected as the experimental SPTBL. The boundary layer was fully developed into turbulence in the test section. He *et al.* used particle image velocity (PIV) technique to study this TBL in detail. The relevant parameters are shown in Table 2 [22], where δ^* and θ were the compressible displacement thickness and momentum thickness, respectively. In order to facilitate the writing of the article, we abbreviated the boundary layer thickness δ_{99} as δ . The flow direction was from left to right, as illustrated in Fig. 2. The direction of flow was the positive direction of the x -axis, the z -axis was positive in the upward direction of the lower wall of the vertical wind tunnel, and the direction of the y -axis was determined by the right-hand rule.

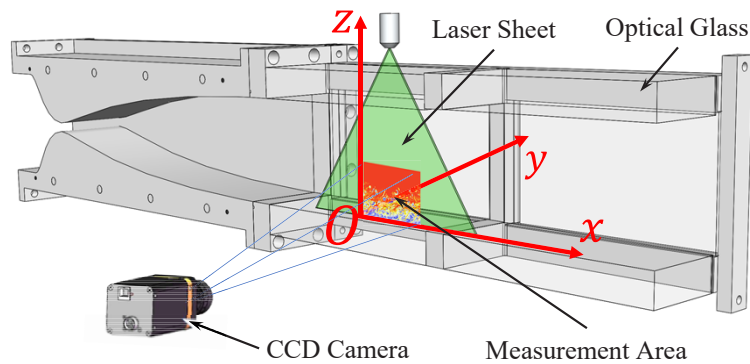


Fig. 2. Schematic of experimental arrangement.

Table 2. Parameters of experimental SPTBL

δ/mm	δ_*/mm	θ/mm	δ_*/δ	θ/δ	$H = \delta_*/\theta$	Re_θ	Re_δ	$u_\tau/(\text{m} \cdot \text{s}^{-1})$	C_f
10.2	3.66	0.68	0.36	0.067	5.38	5100	76500	29.9	1.70×10^{-3}

2.3. Density measurement based on NPLS

In this paper, the instantaneous density field of SPTBL was measured by NPLS technique. NPLS technique utilized titanium dioxide particles with a nominal particle size of 5nm as the tracer particle, distributed them into the air at the entrance of the wind tunnel evenly, and used a double-cavity laser to generate an exciting light source with a wavelength of $\lambda = 532\text{nm}$, pulse energy of 500mJ, and a pulse time of 6ns, which was transmitted through the light guide arm and converted into a laser sheet with a thickness of 0.5mm by the sheet lens group to illuminate the measurement area. The light scattered by nano-tracer-particles was received by a charge coupled device (CCD) camera with a resolution of 2048×2048 pixels. The measurement range of the CCD camera used for measurement in the $x \times z$ plane was $2\delta \times 2\delta$, and the spatial resolution, $\Delta = 11.4\mu\text{m/px}$.

The key to obtaining the supersonic flow density based on NPLS technique was to make the scattering light of the nano-tracer-particle correctly reflect the density information. The maximum Stokes number of the tracer example was obtained through the oblique shock response measurement by Zhao *et al.* [23], $St_{max} = 0.015 \leq 0.05$, which met the requirements of the flow following criteria proposed by Samimy and Lele [24]. It showed that particles had good following performance in supersonic flow, which could meet the requirements of obtaining density field and reflecting the fine flow structure. The density field measurement method based on NPLS made usage of the good following property of nano-tracer-particles. The concentration of nano-tracer-particles was proportional to the local flow density, and it determined the local light intensity of Rayleigh scattering. Therefore, the local density field was a function of the local gray scale of the NPLS image. Via using the shock wave to calibrate the density field measurement, the relationship between the local gray scale and the local flow density could be quantified. At last, the measurement of the instantaneous density field of a certain section could be realized [25]. Further information on NPLS technique and its application in density measurement could be found in the Ref. [19] and [26].

2.4. Ray-tracing method

Based on the density field measured above, the light propagation path and optical path length (OPL) could be calculated by ray tracing. According to the Gladstone-Dale equation, the refractive index field, n , could be obtained from the density field as

$$n = 1 + \rho K_{GD}(\lambda), \quad (1)$$

where K_{GD} was the Gladstone-Dale constant, which is related to the wavelength λ . When the light wavelength, $\lambda = 532\text{nm}$, $K_{GD} \approx 2.22 \times 10^{-4}\text{m}^3/\text{kg}$.

When light propagated in a continuous refractive index field, its trajectory followed the ray equation,

$$\frac{d}{ds} \left(n \cdot \frac{d\mathbf{r}}{ds} \right) = \nabla n, \quad (2)$$

where $\mathbf{r}$ was the position vector of the point on the ray trajectory; s was the distance of the light propagation. Due to the non-uniform and non-steady characteristics of compressible turbulence, the analytical solution of equation could not be obtained. In this paper, the light propagation path was obtained by Runge-Kutta method with third-order accuracy, and the relative error was less than 7.5×10^{-4} [27,28].

The OPL was obtained by integrating the light propagation path,

$$OPL = \int_L n dl, \quad (3)$$

where L is the light transmission path.

2.5. Multi-scale analysis based on curvelet transform

The aero-optical effects were essentially caused by the density fluctuation along light transmission path. The curvelet transform could quantify the geometric characteristics scales and preferential orientations of turbulent vortex in the flow field [29–31].

For a two-dimensional scalar (density) field $\varphi \in \mathbb{R}^2$, the Fourier transform was performed to obtain the frequency domain,

$$\hat{\varphi}(\mathbf{k}) = \frac{1}{2\pi} \int_{\mathbb{R}} \varphi(\mathbf{x}) e^{-i\mathbf{k} \cdot \mathbf{x}} d\mathbf{x}, \quad (4)$$

where $\mathbf{x} = (x, z)$ and $\mathbf{k} = (k_1, k_2)$.

To extract the frequency components of scale j and direction l ($l = -2^{[j/2]}, \dots, 2^{[j/2]} - 1$) from the frequency space $\hat{\varphi}(\mathbf{k})$, the frequency window function was defined as,

$$U_j(r, \xi) = 2^{-3j/4} W(2^{-j}r) V_l(\xi), \quad (5)$$

where $j \geq j_0$, $r = \sqrt{k_1^2 + k_2^2}$, $\xi = \tan^{-1} k_2/k_1$. $W(r)$ was the “radial window” function. $V(\xi)$ was the “angular window” function. The definition of the window function could be found in Ref. [29]. It could be seen from the definition that U_j was a wedge-shaped window in polar coordinates.

Via applying the radial window $W(r)$ in the frequency window function, the geometric characteristics of the two-dimensional density field could be extracted at the characteristic length scale $\mathcal{L}_j = 2^{-j}$, $j \in \mathbb{N}_0$. The normalized turbulent length scale decomposition after filtering was shown in Table 3. For each scale j , the original scalar field $\varphi(\mathbf{x})$ was decomposed into multi-scales,

$$\varphi_j(\mathbf{x}) = \int \hat{\varphi}(\mathbf{k}) W(2^{-j}r) e^{i\mathbf{k} \cdot \mathbf{x}} d\mathbf{k}, \quad (6)$$

where $\varphi_j(\mathbf{x})$ was the component field after filtering at the scale j .

Table 3. Characteristic length scale $\mathcal{L}_j^a$

Scale(j)	1	2	3	4	5	6	7
$\mathcal{L}_j/\delta$	5	2.5	1.25	0.625	0.313	0.156	0.078
$\mathcal{L}_j/\theta$	75	37.5	18.75	9.375	4.688	2.344	1.172
$\mathcal{L}_j/l_c$	289.5	144.8	72.38	36.19	18.09	9.047	4.523
$\mathcal{L}_j/\eta$	1110	554.9	277.4	138.7	69.36	34.68	17.34
$\mathcal{L}_j/\Delta$	4474	2237	1118	559.2	279.6	139.8	69.90

^aNormalized by the velocity boundary layer thickness δ , the moment boundary layer thickness θ , the smallest optically active scale l_c , the Kolmogorov scale η and the spatial resolution of the test image Δ .

3. Experiment results and discussion

3.1. Verification of wavefront measurement results

The instantaneous density field result of the SPTBL measured by experiment was shown in Fig. 3. In total, 100 sets of instantaneous density fields were measured and utilized for further investigations. The measurement range spans $x/\delta \in [0, 2]$, $z/\delta \in [0, 1.5]$, and the spatial resolution was described by the pixel size Δ , and $\Delta = 11.4\mu\text{m}$.

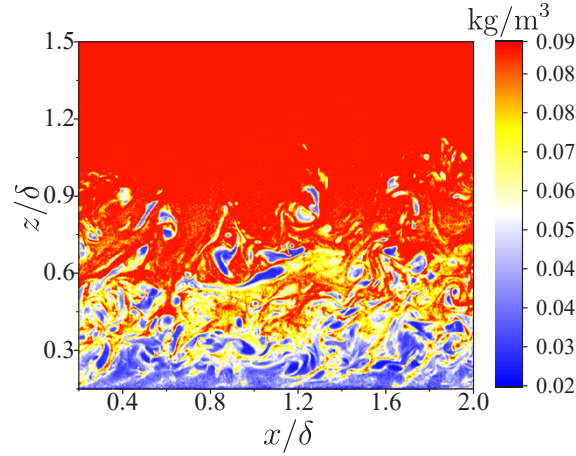


Fig. 3. Instantaneous result of density field of SPTBL obtained using NPLS.

The incident beam whose intensity meet a binary distribution was selected as the statistical object, and its amplitude distribution in one-dimensional spatial distribution meet

$$A(x) = \begin{cases} 1 & , \quad |x - \delta| \leq Ap/2 \\ 0 & , \quad |x - \delta| > Ap/2 \end{cases} \quad (7)$$

where, Ap was the diameter of the optical aperture of the binary beam. Different binary beams with different Ap were selected as research objects, and the distortion situation was statistically with different Ap . The ray-tracing method was used to numerically solve the trajectory of the 90° incident light and obtain its OPL .

The OPD within the beam was statistically expressed as

$$OPD(x) = OPL(x) - [OPL(x)], \quad (8)$$

where the square bracket denoted taking the spatial average over the optical aperture of the beam.

We noted that the starting point of OPD was the average value of OPL inside the beam. This led to different starting points for the OPD calculation for beams with different optical aperture sizes in the same instantaneous flow field. In previous studies, the influence of optical apertures was not involved, and the average value of OPL over the entire measurement range was generally selected as the starting point for the calculation of OPD . Since the measurement range was fixed, there was no problem with the definition of the starting point. In this study, in order to compare the influence of the starting point of OPD , we also took the average value of OPL over the entire field as the starting point, and defined $\widetilde{OPD}$ as

$$\widetilde{OPD}(x) = OPL(x) - \langle OPL(x) \rangle, \quad (9)$$

where the angled brackets denoted the spatial average of the full field.

OPD_{rms} , the spatial RMS value of OPD , was commonly used as an evaluation index for the aero-optical effects. Within the range of the optical aperture Ap , we calculated the spatial RMS of OPD using two different starting points as follows:

$$OPD_{rms} = RMS(OPD), \quad (10)$$

$$\widetilde{OPD}_{rms} = RMS(\widetilde{OPD}). \quad (11)$$

We calculated OPD_{rms} within different optical aperture sizes and $\widetilde{OPD}_{\text{rms}}$ over the entire field. The average and standard deviation of 100 instantaneous results were calculated and the error bars were plotted in Fig. 4. The momentum thickness θ was selected for dimensionless transformation of Ap . In Fig. 4, The red solid line represents OPD_{rms} defined in Eq. (10), and the blue dashed line represents $\widetilde{OPD}_{\text{rms}}$ defined in Eq. (11).

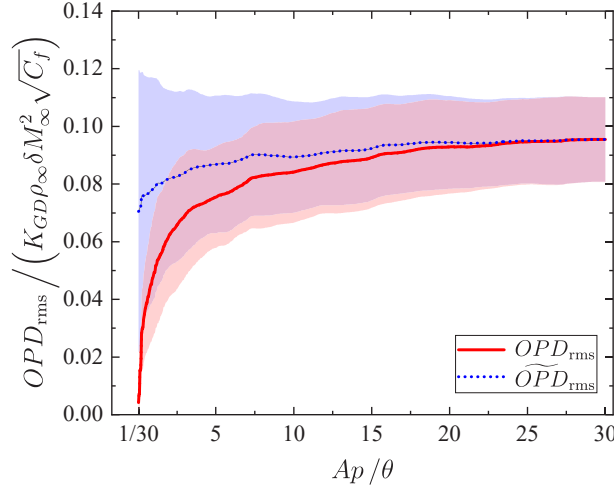


Fig. 4. OPD_{rms} affected by the optical aperture size.

The OPD_{rms} statistical results were compared with theoretical predictions of the scaling law proposed by Wyckham and Smits [13],

$$OPD_{\text{rms}} = C_W K_{\text{GD}} \rho_{\infty} \delta M_{\infty}^2 \sqrt{C_f} F(M_{\infty}), \quad (12)$$

$$F(M_{\infty}) = (1 + M_{\infty}^2(\gamma - 1)[1 - r(U_C/U_{\infty})^2]/2)^{-3/2}. \quad (13)$$

The boundary layer thickness δ , the mainstream density ρ_{∞} , the mainstream Mach number M_{∞} , and the surface friction coefficient C_f were given in Table 2. Gordeyev and Rennie [15] further set the model parameter C_W to 0.2 and the total temperature recovery coefficient for adiabatic wall to $r = 0.89$. The air adiabatic ratio γ was set to $\gamma = 1.4$. The mainstream velocity U_{∞} and the convection velocity of the large-scale optical structure U_C were set to $U_C = 0.84U_{\infty}$. When $M_{\infty} = 3.0$, the theoretical prediction given by the scaling law was $OPD_{\text{rms}} / (K_{\text{GD}} \rho_{\infty} \delta M_{\infty}^2 \sqrt{C_f}) = C_W F(M_{\infty}) = 0.09270$, which was in good agreement with the experimental data in Fig. 4 for large optical aperture size.

Due to the different starting points selections of OPD , OPD_{rms} and $\widetilde{OPD}_{\text{rms}}$ showed a significant difference at the smaller optical aperture, as shown in Fig. 4. with the increase of optical aperture size, the two curves approached to each other, which was determined by the definition of OPD . It should be noted that, although the mean of 100 fields decreased with the decrease of optical aperture size, the standard deviation of $\widetilde{OPD}_{\text{rms}}$ increased. Therefore, the mean of $\widetilde{OPD}_{\text{rms}}$ was not reliable as a statistic. In the following parts, we would continue to select OPD_{rms} as the statistic for analysis.

3.2. Beam distortion at different optical apertures

OPD_{rms} was used to quantify the aero-optical effects inside the beam. In addition, after passing through the turbulent boundary layer, the optical beam center offset $\bar{x}$ and the relative spread of

the beam center $\overline{x'^2}$ could be defined as follows [32]:

$$\bar{x} = \overline{OPD_x}|_{z_1}(z - z_1) + \bar{x}|_{z_1} \equiv \alpha(z - z_1) + \bar{x}|_{z_1}, \quad (14)$$

$$\begin{aligned} \overline{x'^2} &= \overline{OPD_{x'}^2}|_{z_1}(z - z_1)^2 + 2x'\overline{OPD_{x'}}|_{z_1}(z - z_1) + \overline{x'^2}|_{z_1} \\ &\equiv \beta(z - z_1)^2 + 2\gamma(z - z_1) + \overline{x'^2}|_{z_1} \end{aligned} \quad (15)$$

In the above equations, the overline denoted the weighted average according to the intensity I .

$$\overline{f(x)} = \frac{\int f(x)I(x)dx}{\int f(x)dx} \quad (16)$$

$$I(x) = |A(x)|^2 \quad (17)$$

And the subscript x denoted the derivative with respect to x , such as $OPD_x = dOPD(x)/dx$, and the prime denoted the offset relative to the mean, such as $x' = x - \bar{x}$, $OPD_{x'} = OPD_x - \overline{OPD_x}$.

The average of 100 fields was taken as the statistical result of the center offset $\bar{x}$ and beam spread $\overline{x'^2}$. The standard deviation $s_{\bar{x}}$ of the 100 center offset $\bar{x}$ was used to measure the beam center jitter.

For $z - z_1 = 10\text{km}$, the optical distortion at different optical aperture diameter (Ap) was shown in Fig. 5. With Ap increased, OPD_{rms} also increased synchronously. Although OPD_{rms} was used to measure the intensity of the aero-optical effects, OPD_{rms} could not describe the degree of aero-optical distortion comprehensively, as shown in Fig. 5. With increase of Ap , the jitter of the beam ($s_{\bar{x}}$) was reduced, while the center offset ($\bar{x}$) and beam spread ($\overline{x'^2}$) experienced a sharp rise at small apertures ($<\theta/5$), and they were also suppressed at larger aperture sizes ($>\theta/2$). To fully characterize the beam distortion, more intuitive statistical parameters, e.g. jitter ($s_{\bar{x}}$), offset ($\bar{x}$), and beam spread ($\overline{x'^2}$), should be taken into account.

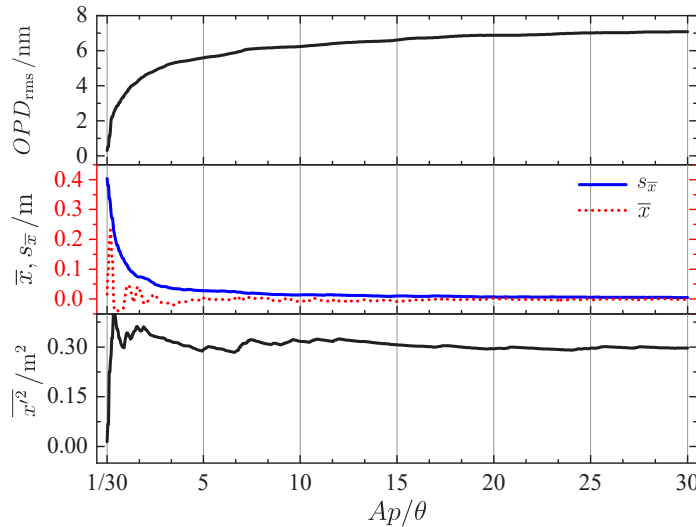


Fig. 5. Beam distortion affected by the optical aperture.

The aero-optical distortion was closely related to the scale of the vortex structure. Varying scales vortex structures could lead to three distinct levels of beam distortion [33]: small-scale vortex structures with a wavelength scale would induce energy scattering and result in Strehl

ratio loss; medium-scale vortex structures, which were larger than the wavelength but smaller than the optical aperture, would cause the beam spread about center and result in imaging blur; finally, large-scale vortex structures, which were larger than the optical aperture, would lead to beam jitter and offset. The larger the optical aperture size was, the smaller the proportion of turbulence structure larger than the optical aperture size was, and the jitter and offset of the beam would be suppressed. In the following research, it would be more intuitively reflected.

Based on the influence of turbulent structure scale, Mani *et al.* developed a theory to estimate the smallest optically active scale in a general aero-optical framework [12].

$$l_c \approx \frac{\xi^{3/7}}{M^{12/7}(n_\infty - 1)^{6/7}} \frac{\lambda^{6/7} l^{4/7}}{z_1^{3/7}}, \quad (18)$$

where l_c was the smallest optically active scale. According to their analysis, in the limit of high Reynolds number, the smallest optically active scale (l_c) did not depend on the Kolmogorov scale, but only depends on the characteristic turbulent Mach number (M), the free stream refractive index (n_∞), optical wavelength (λ). And l was the length scale of the largest turbulent vortex and z_1 was the turbulent field depth in the direction of beam propagation, ξ was the error threshold. When $\xi \leq 5\%$, the accuracy was 95%. For the experimental boundary layer in this paper, $l_c = 176.2\mu\text{m} = 0.017\delta = 15.5\Delta$. In Fig. 5, $\bar{x}$ and $\bar{x}^2$ were highest when Ap was slightly larger than l_c . Next, we would analyze the turbulent structure near this scale to explore the cause of the peak in further.

3.3. Beam distortion at different turbulence scales

According to the method in section 2.5, the density distributions of the SPTBL at different scales extracted by curvelet transformation were shown in Fig. 6. The characteristic length of each turbulence structure scale was given in Table 3. Although curvelet transformation could be applied to avoid the appearance of residual field caused by the decomposition method, due to the existence of image boundary, there would be large discontinuous error at the image edge. In order to suppress the error, replication continuation was used in the images before decomposition.

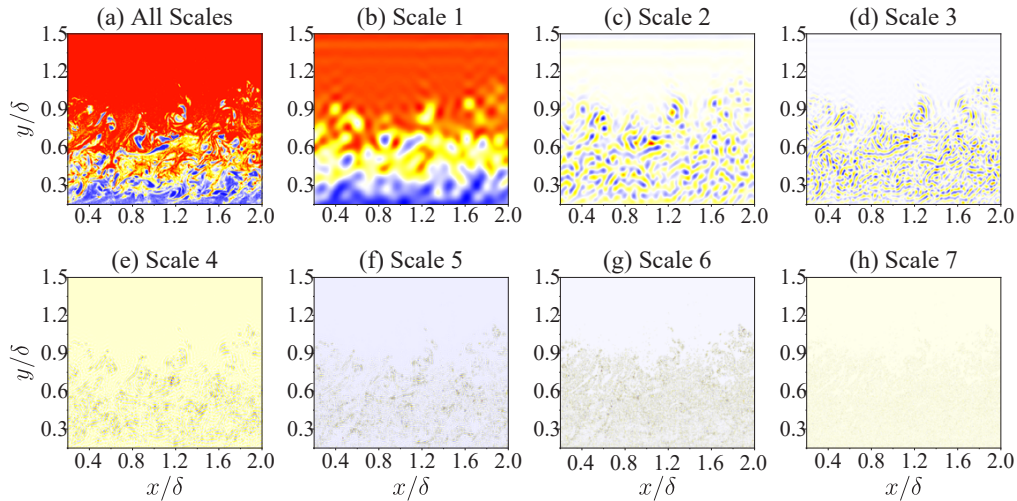


Fig. 6. Curvelet (multiscale geometric) decomposition results of the density field of SPTBL.

The influence of the turbulent structure scales on aero-optical distortion over different optical aperture sizes was statistically analyzed. As shown in Fig. 7, to facilitate the analysis of the

characteristics of the smaller optical aperture ($<\theta$), the logarithmic coordinate system was adopted for the horizontal axis. In order to verify the effectiveness of multi-scale decomposition, the aero-optical distortion caused by the flow field at each scale was superimposed and compared with that caused by the full-scale flow field. Due to the definitions of $s_{\bar{x}}$ and OPD_{rms} , when they were superimposed, the individual results should be squared and then summed before taking the square root, as indicated by the red dashed line in Fig. 7. On the other hand, when $\bar{x}$ and x'^2 were superimposed, they could be linearly summed directly, as indicated by the red solid line in Fig. 7.

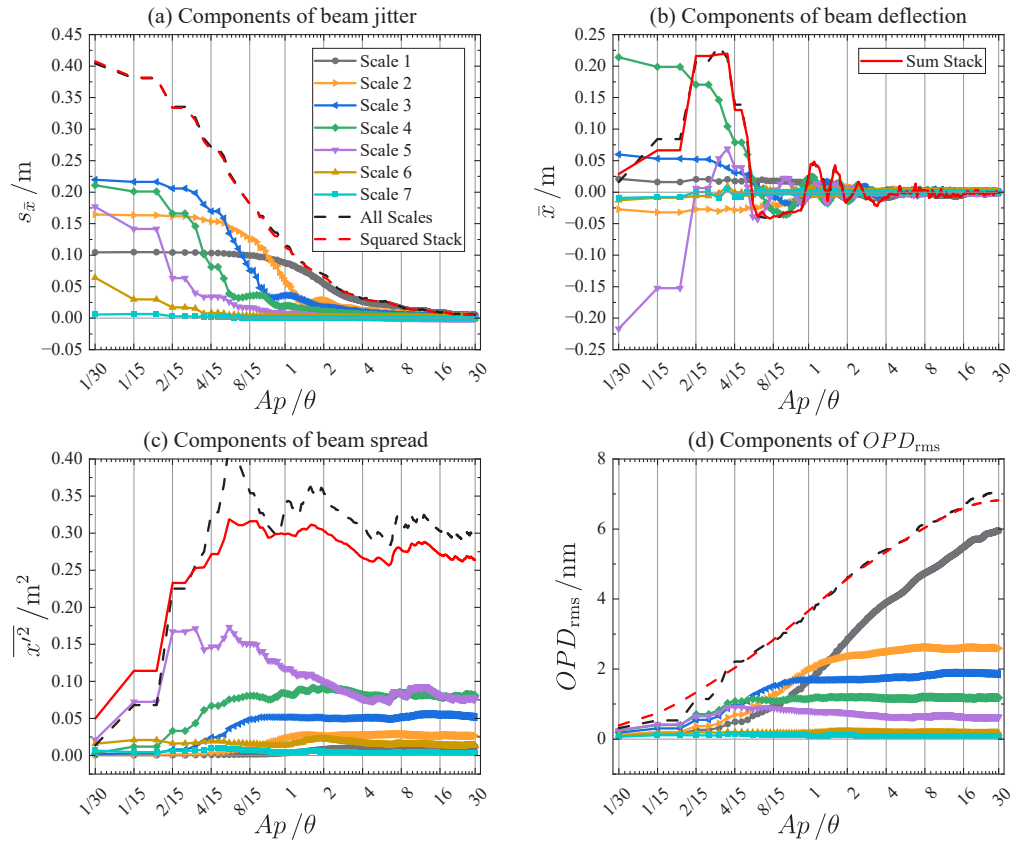


Fig. 7. Beam aberration at different scales.

Since $s_{\bar{x}}$ and $\bar{x}$ were mainly caused by turbulent structures larger than the optical aperture, $s_{\bar{x}}$ and $\bar{x}$ at each scale were suppressed with the increase of the optical aperture size. The smaller the characteristic scale of the turbulent flow, the easier the suppression effect was reflected. In Fig. 7(a), it could be observed that $s_{\bar{x}}$ caused by scale 5 decreased as Ap increased. The reduction in $s_{\bar{x}}$ was relatively slow between $\theta/30$ and $\theta/15$, but became more rapid between $\theta/15$ and $2\theta/15$. Near $4\theta/15$, the reduction in $s_{\bar{x}}$ became slow again. A similar pattern was observed for scale 4. The characteristic length scale of scale 4 was twice that of scale 5, and the optical aperture size where the transition occurred happened to also be twice that of scale 5. Similar doubling relations were observed among the other scales as well. These findings suggested that the same flow field could induce different aero-optical distortions at different optical aperture sizes, which was indeed caused by the presence of specific turbulent structures in the flow field. In this study, we referred to this phenomenon as the coupling effect between optical aperture sizes and turbulent scales.

In addition to $s_{\bar{x}}$, both $\bar{x}$ and $\overline{x'^2}$ also exhibited coupling effects. Since $\overline{x'^2}$ was caused by turbulent structures smaller than the optical aperture size, as the optical aperture size increased, the proportion of flow scales smaller than the optical aperture size also increased, which lead to an increase of $\overline{x'^2}$ until almost all turbulent structures were smaller than the optical aperture size and the growth of $\overline{x'^2}$ almost stopped. However, there were some differences in $\overline{x'^2}$ generated by the flow field of scale 5, and it would be analyzed in the following parts. From the above results, it could be found that due to the coupling effects of between optical aperture sizes and turbulent scales, the proportion of turbulence with different scales to aero-optical distortion would be different under different optical apertures. In order to quantitatively analyze it, the results of $s_{\bar{x}}$ and $\overline{x'^2}$ were selected. They were meaningful for engineering research. As shown in Fig. 8, they were plotted as a stacked graph.

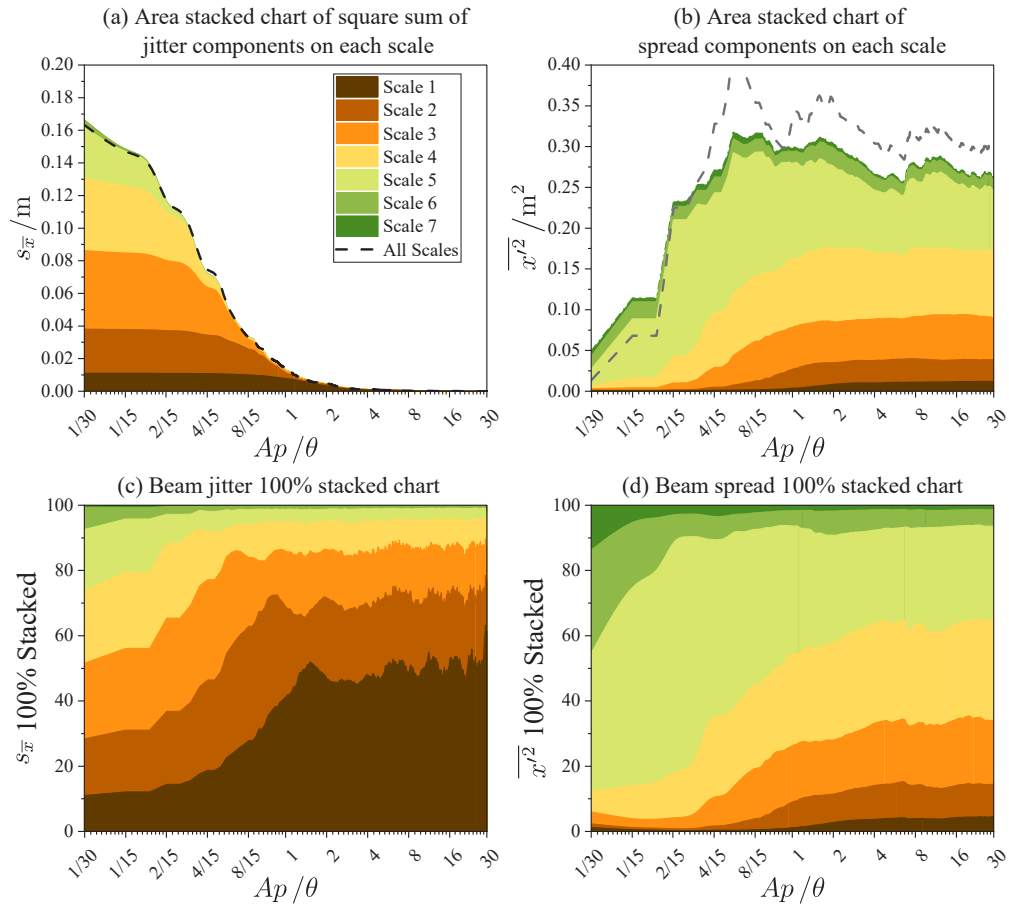


Fig. 8. Beam jitter and spread ratio at different scales.

Figure 8(a) was the area stacked chart of $s_{\bar{x}}$ on each scale. $s_{\bar{x}}$ was mainly produced by turbulent structures larger than the optical aperture. Therefore, with the increase of Ap , the turbulent structures larger than Ap became fewer, and $s_{\bar{x}}$ was continuously suppressed. Figure 8(c) was the 100% stacked chart of $s_{\bar{x}}$ on each scale. With the increase of Ap , the contribution of small-scale (scale 5, 6, 7) turbulent structures to $s_{\bar{x}}$ decreased significantly. At a larger optical aperture ($>\theta$), the largest turbulent structure scale (scale 1) was the major influencing factor. And the

large-scale (scale 1, 2, 3) turbulence structures with characteristic scale larger than δ made the main contribution to $s_{\bar{x}}$. Figure 8(b) was the area stacked chart of $\overline{x'^2}$ under the turbulent flow field of each scale. It could be seen that $\overline{x'^2}$ on each scale was increasing and tending to be stable, but scale 5 begun to decrease after reaching the maximum value. Figure 8(d) showed the 100% stacked chart of $\overline{x'^2}$, which showed the contribution of each scale turbulent structure to $\overline{x'^2}$. The intensity of $\overline{x'^2}$ was related to the coupling effect and the intensity of the turbulent field. The aero-optical intensity corresponding to scale 6 and 7 were much smaller than that corresponding to scale 5, so $\overline{x'^2}$ dominated by small-scale turbulent structures was dominated by the turbulent structures with scale 5. At the same time, the smallest turbulent scale (η) had little impact on $\overline{x'^2}$.

The turbulent structures with scale 5 showed some special characteristics. In Fig. 7(b), when $Ap \leq \theta/2$, $\bar{x}$ corresponding to scale 5 and scale 4 were relatively larger than $\bar{x}$ corresponding to other scales. Different from other scales, scale 5 had a larger negative offset. Due to the coupling effect, the offset corresponding to scale 5 was suppressed faster than that of scale 4, thus leading to the peak in the total scale offset at $Ap \approx \theta/2$. For $\overline{x'^2}$, scale 5 played a dominant role and would show a suppression phenomenon of $\overline{x'^2}$ with $Ap > \theta/2$. The optical aperture size of turning point $\theta/2 \approx l_c$ matched the estimation of the smallest optically active scale (l_c), and the turbulent structures slightly larger than l_c need to be studied in further.

4. Conclusion

Based on curvelet transform method, the density field of specific turbulence scale was extracted. The influence of optical aperture on aero-optical effects induced by SPTBL was investigated. It was verified that $s_{\bar{x}}$ and $\bar{x}$ were mainly caused by turbulence structures larger than Ap , and $\overline{x'^2}$ was mainly caused by turbulence structures smaller than Ap . It was found that the large-scale turbulence structures with characteristic scale larger than δ made the main contribution to $s_{\bar{x}}$ and $\bar{x}$, while the turbulence structures with relatively small scale (0.313δ) made the main contribution to $\overline{x'^2}$. The coupling effect of optical aperture and turbulent structure on aero-optical effects was clarified. With the increase of Ap , $s_{\bar{x}}$ could be effectively suppressed. The turbulence structures with characteristic optical scale slightly larger than the optical active scale ($\approx l_c$) had some different influence on $\bar{x}$ and $\overline{x'^2}$, which were worth to be studied in further.

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